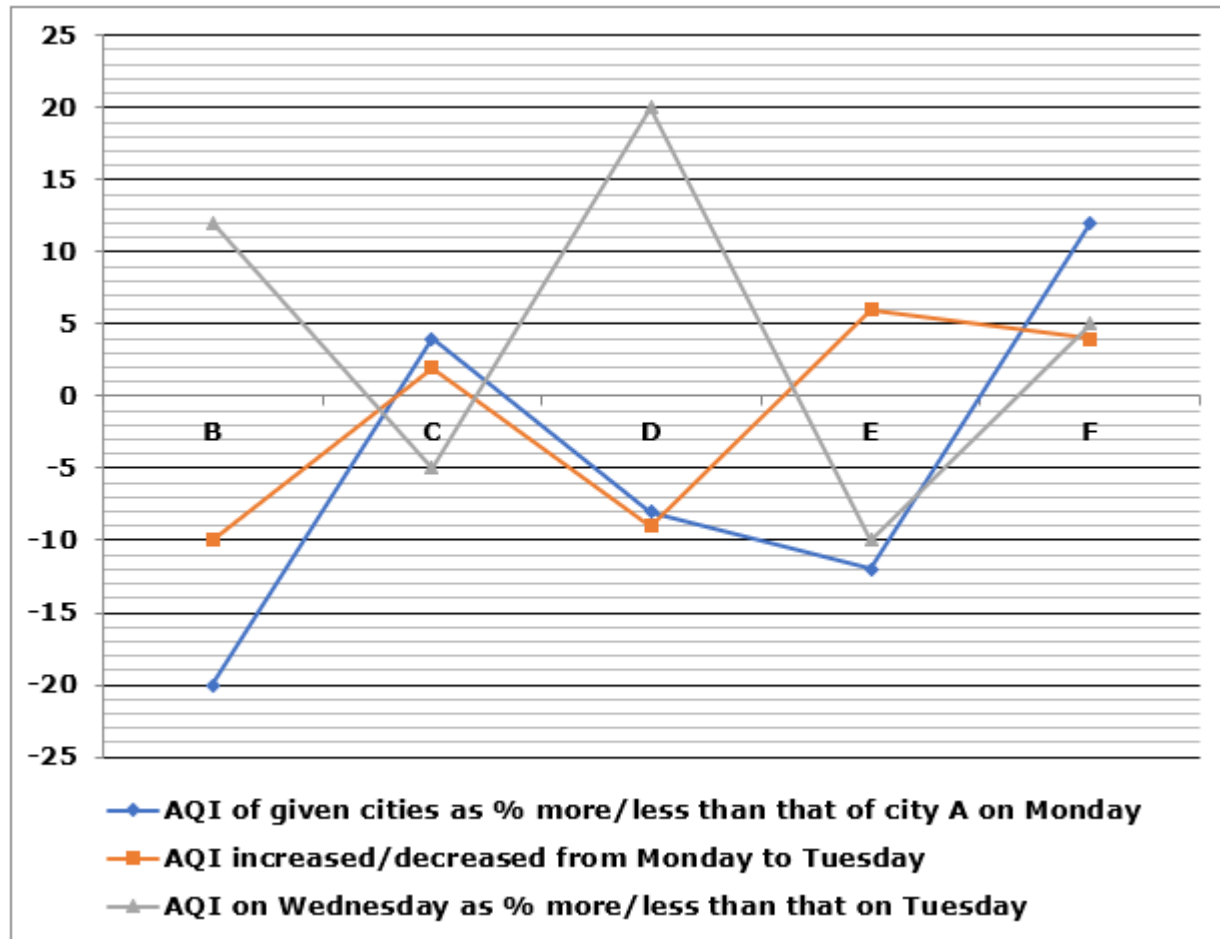


1. Questions

Study the following data carefully and answer the questions:

The given line graph shows the AQI (Air quality index) of six cities i.e. A, B, C, D, E, and F on three different days Monday, Tuesday, and Wednesday.



Note:

I). The average AQI of each of the given cities on Monday was 72.

The AQI of city B is decreased by $X\%$ from Monday to Tuesday. If the difference between the AQI's of cities A and C on Tuesday is 10 and the AQI of city A on Tuesday is a multiple of 7, then by what percentage has the AQI of city A increased/decreased on Tuesday compared to Monday?

- $[X - 10]\%$
- $[X - 16(2/3)]\%$
- $[X - 6(2/3)]\%$
- None of these
- $[X - 6]\%$

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2. Questions

The AQI of city A on Wednesday is $2(2/3)\%$ more than that on Monday. On Wednesday, find the average AQI of those cities (excluding city A), whose AQI is better than that of city A? (The city

which has a lower AQI is considered a better AQI)

- a. 63.4
- b. 72.1
- c. 65.5
- d. 67.2
- e. 74.3

3. Questions

In city D, if the average AQI for the whole week is 65, then find the ratio of the average AQI over the given three days to the average AQI over the remaining four days of the week?

- a. Can't be determined
- b. 67: 61
- c. 134: 127
- d. 34: 27
- e. 4: 3

4. Questions

The ratio of the 2nd lowest AQI of city on Monday to the highest AQI of city on Monday is M: N. Which of the following statements is/are true?

I: The value of $(M * N)$ is a perfect square.

II: The value of $(15N - 5M)$ is a perfect cube.

III: The value of $(8M + 5N)$ is either a perfect square or a perfect cube.

- a. None is true
- b. Only I and II
- c. All are true
- d. Only I and III
- e. Only II and III

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5. Questions

If the AQI of city A on Monday decreased by $33\frac{1}{3}\%$, then the AQI of city D on Wednesday would have been what percent of the AQI of city E on Wednesday?

- a. $88\frac{2}{3}\%$
- b. $92\frac{2}{3}\%$

- c. $76\frac{2}{3}\%$
- d. $84\frac{2}{3}\%$
- e. $98\frac{2}{3}\%$

6. Questions

Quantity I: If the average AQI for whole week in city C is 75, then find the average AQI over the remaining four days in city C?

Quantity II: Find the average AQI of cities B, C, D, E and F on Tuesday?

- a. Quantity I < Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity I $\leq$ Quantity II
- d. Quantity I > Quantity II
- e. Quantity I = Quantity II or relation can't be established

7. Questions

Study the following data carefully and answer the questions:

The data given below is related to the initial investment made by A, B, C and D in a partnership.

The difference between the initial investments of A and B is ₹3000 and the ratio of the initial investment of B to that of C is 9: 5. The initial investment of D is ₹4000 more than that of C. The ratio of the initial investment of A to that of D is 15: 14 and the initial investment of A is more than ₹7500.

If after 't' months, A withdrew 40% of his initial investment, after 2 more months, B withdrew $33\frac{1}{3}\%$ of his initial investment and at the end of 8th month, the ratio of the profit shares of A to B is 8: 11, then find the time period for B withdraw $33\frac{1}{3}\%$ of his initial investment?

- a. Can't be determined
- b. 5 months
- c. 7 months
- d. 4 months
- e. 6 months

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8. Questions

The ratio of initial investment of 'B' to 'D' is M: N. If the total profit received by B and D together after 12 months, were ₹[(M + N) * 5M * 100], then find the difference between the profit shares of B and D?

- a. ₹7200
- b. ₹4500

- c. ₹9000
- d. ₹6300
- e. ₹6000

9. Questions

The difference between, B's profit share after 4 months and C's profit share after 10 months is ₹10500. If the total profit received by B after 4 months and that received by C after 10 months, were invested in a scheme together, which gives simple interest at 7.5% rate, then find the sum of the interest received by B and C together after 2 years?

- a. ₹9255
- b. ₹9675
- c. ₹9425
- d. ₹9515
- e. ₹9365

10. Questions

If the time periods for the investments of A, B, C and D are 8 months, 10 months, 9 months and 15 months respectively and the total profit received by A, B, C and D together is ₹35000, then which of the following statements is/are not true?

I: The difference between the profit shares of B and C is ₹5850.

II: The difference between the profit shares of B and D is completely divisible by 140.

III: The total profit share of A and C together is a perfect square.

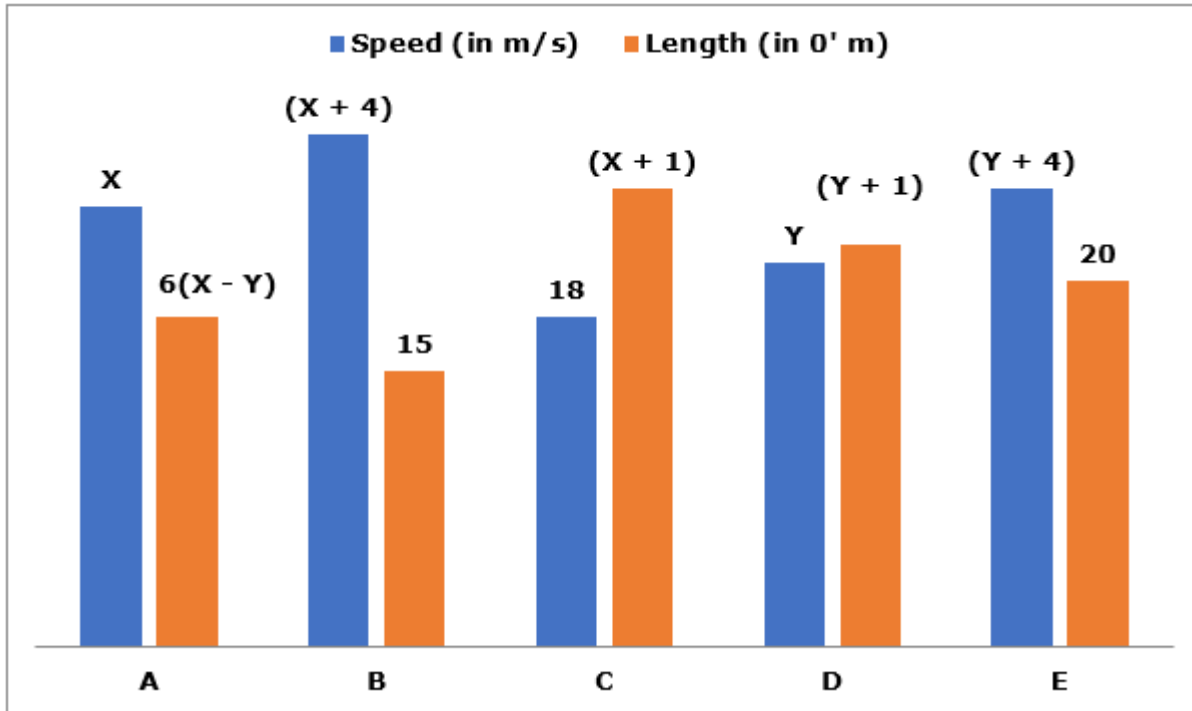
- a. Only II and III
- b. None is true
- c. Only III
- d. Only I and II
- e. Only II

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11. Questions

Study the following data carefully and answer the questions:

The given bar graph shows below is related to the speeds (in m/s) and lengths (in m) of five trains i.e. A, B, C, D and E.



Notes:

I). The time taken by train A to cross a long bridge whose length is 228 m in 17 seconds.

II). The length of train D is 88% of that of train C.

When the trains A and B are 450 m away from each other and coming towards each other, they will cross each other in 'x' seconds. When the train B is 150 m behind the train A, then the train B will cross the train A in 'y' seconds. Find the ratio of 'x' to 'y'?

- a. 1:12
- b. 3:8
- c. 1:8
- d. 3:10
- e. 1:3

12. Questions

If the train D crosses a long tunnel whose length is 'd' m in 15 seconds and the train E crosses a long tunnel whose length is $(d + 5)$ m in 't' seconds, then find the time taken by train C to cross a long tunnel whose length is $(2d - 8)$ m?

- a. $(t + 5)$ seconds
- b. '3t' seconds
- c. $(t - 2)$ seconds
- d. $(t + 10)$ seconds
- e. '2t' seconds

13. Questions

Which of the following statement(s) is/are not true?

I: The time taken by train B to cross a long platform of length 186 min 14 seconds.

II: The ratio of the length of train A to that of train E is 9: 10.

III: The average speed of trains A, C, and D is 75.6 km/h.

- a. Only I and II
- b. Only III
- c. Only II and III
- d. Only II
- e. Only I

14. Questions

The train B crosses train C in 'p' seconds while running in the same direction and the train E crosses train D in 'q' seconds while running in the same direction. Which of the following relations between 'p' and 'q' is correct?

- a. $21p = 10q$
- b. $2p + q = 185$
- c. $p * q = 4200$
- d. Both (a) and (b)
- e. Both (b) and (c)

15. Questions

If the average speed of trains C, D, E, and F is 72 km/h and the average length of trains A, B, C, and F is 175 m, then find the time taken by train F to cross a long bridge whose length 232 m?

- a. 24 seconds
- b. 22 seconds
- c. 19 seconds
- d. 25 seconds
- e. 18 seconds

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16. Questions

The table given below contains two columns i.e. M and N. Column M contains three different quadratic equations and column N contains three different roots. Match the correct equation with its correct roots.

Column M	Column N
P: $x^2 - 5x - 104 = 0$	I: 15
Q: $y^2 - 9y - 70 = 0$	II: -8
R: $z^2 - 11z - 60 = 0$	III: 14

- P-I, Q-II, R-III
- P-II, Q-III, R-I
- P-III, Q-II, R-I
- P-II, Q-I, R-III
- P-III, Q-I, R-II

17. Questions

Column M	Column N
P: $4x^2 - 29x + 45 = 0$	I: 9/4
Q: $4y^2 + 7y - 36 = 0$	II: 5
R: $z^2 - z - 20 = 0$	III: -4

- P-I, P-III, Q-I, Q-II, R-II, R-III
- P-I, P-III, Q-II, Q-III, R-I, R-II
- P-II, P-III, Q-I, Q-II, R-I, R-III
- P-II, P-III, Q-I, Q-III, R-I, R-II
- P-I, P-II, Q-I, Q-III, R-II, R-III

18. Questions

Column M	Column N
P: $3x^2 - 25x + 52 = 0$	I: 4
Q: $3y^2 - 8y - 16 = 0$	II: 4/3
R: $3z^2 + 8z - 16 = 0$	III: -4

- P-II, Q-I
- Q-II, R-I
- P-I, R-III
- Q-III
- P-III

19. Questions

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Column M	Column N
P: $x^2 - x - 42 = 0$	I: 17
Q: $y^2 - 25y + 136 = 0$	II: -6
R: $z^2 - 11z - 102 = 0$	III: 8

- a. P-II, Q-III, R-I
- b. P-II, Q-II
- c. P-III
- d. Q-II, R-III
- e. Q-I, R-III

20. Questions

Column M	Column N
P: $2x^2 - 47x - 5A = 0$	I: 18
Q: $2y^2 - 31y - 6A = 0$	II: 24
R: $3z^2 - 67z - 8A = 0$	III: 25

One root of 'x' is (-3/2).

- a. P-I, Q-II, R-III
- b. P-III, Q-II, R-I
- c. P-III, Q-I, R-II
- d. P-II, Q-III, R-I
- e. P-II, Q-I, R-III

Explanations:

1. Questions

Let the AQI of city A on Monday be 100a

On Monday,

The AQI's of city B = $100a * [1 - (20/100)] = 100a * (80/100) = 80a$

The AQI's of city C = $100a * [1 + (4/100)] = 100a * (104/100) = 104a$

The AQI's of city D = $100a * [1 - (8/100)] = 100a * (92/100) = 92a$

The AQI's of city E = $100a * [1 - (12/100)] = 100a * (88/100) = 88a$

The AQI's of city F = $100a * [1 + (12/100)] = 100a * (112/100) = 112a$

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According to the question,

$$100a + 80a + 104a + 92a + 88a + 112a = 6 * 72$$

$$576a = 432$$

$$a = 0.75$$

Cities	AQI on Monday	AQI on Tuesday	AQI on Wednesday
A	$100 * 0.75 = 75$	—	—
B	$80 * 0.75 = 60$	$60 - 10 = 50$	112% of 50 = 56
C	$104 * 0.75 = 78$	$78 + 2 = 80$	95% of 80 = 76
D	$92 * 0.75 = 69$	$69 - 9 = 60$	120% of 60 = 72
E	$88 * 0.75 = 66$	$66 + 6 = 72$	90% of 72 = 64.8
F	$112 * 0.75 = 84$	$84 + 4 = 88$	105% of 88 = 92.4

Answer: A

Since, the AQI of city B on Monday = 60

And the AQI of city B on Tuesday = 50

$$\text{So, } X = [(60 - 50)/60] * 100 = 16(2/3)\%$$

Since, the difference between the AQIs of cities A and C on Tuesday = 10

And the AQI of city C in Tuesday = 80

So, the AQI of city A on Tuesday will either be $80 - 10 = 70$ or $80 + 10 = 90$.

Since, the AQI of city A on Tuesday is a multiple of 7.

So, the AQI of city A on Tuesday = 70

Since, the AQI of city A on Monday = 75

$$\text{So, the required percentage} = [(75 - 70)/75] * 100 = 6(2/3)\%$$

Here, checks option wise

Option a) $(X - 10)\% = 16(2/3)\% - 10\% = 6(2/3)\%$ is correct

Hence, option A

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2. Questions

Let the AQI of city A on Monday be $100a$

On Monday,

$$\text{The AQI's of city B} = 100a * [1 - (20/100)] = 100a * (80/100) = 80a$$

$$\text{The AQI's of city C} = 100a * [1 + (4/100)] = 100a * (104/100) = 104a$$

$$\text{The AQI's of city D} = 100a * [1 - (8/100)] = 100a * (92/100) = 92a$$

The AQI's of city E = $100a * [1 - (12/100)] = 100a * (88/100) = 88a$

The AQI's of city F = $100a * [1 + (12/100)] = 100a * (112/100) = 112a$

According to the question,

$$100a + 80a + 104a + 92a + 88a + 112a = 6 * 72$$

$$576a = 432$$

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Cities	AQI on Monday	AQI on Tuesday	AQI on Wednesday
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E	$88 * 0.75 = 66$	$66 + 6 = 72$	90% of 72 = 64.8
F	$112 * 0.75 = 84$	$84 + 4 = 88$	105% of 88 = 92.4

Answer: D

Since, the AQI of city A on Monday = 75

So, the AQI of city A on Wednesday = $\{1 + 2(2/3)/100\} = (308/300)$ of 75 = 77

Since, the AQIs of cities B, C, D, E and F on Wednesday are 56, 76, 72, 64.8 and 92.4 respectively.

So, the AQIs of cities B, C, D and E are lower than that of city A.

And the AQIs of cities B, C, D and E will be better than that of city A.

So, the required average = $(56 + 76 + 72 + 64.8)/4 = 67.2$

3. Questions

Let the AQI of city A on Monday be $100a$

On Monday,

The AQI's of city B = $100a * [1 - (20/100)] = 100a * (80/100) = 80a$

The AQI's of city C = $100a * [1 + (4/100)] = 100a * (104/100) = 104a$

The AQI's of city D = $100a * [1 - (8/100)] = 100a * (92/100) = 92a$

The AQI's of city E = $100a * [1 - (12/100)] = 100a * (88/100) = 88a$

The AQI's of city F = $100a * [1 + (12/100)] = 100a * (112/100) = 112a$

According to the question,

$$100a + 80a + 104a + 92a + 88a + 112a = 6 * 72$$

$$576a = 432$$

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Cities	AQI on Monday	AQI on Tuesday	AQI on Wednesday
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E	$88 * 0.75 = 66$	$66 + 6 = 72$	90% of 72 = 64.8
F	$112 * 0.75 = 84$	$84 + 4 = 88$	105% of 88 = 92.4

Answer: C

The total AQI of city D on all the given three days together = $69 + 60 + 72 = 201$

The total AQI of city D for the whole week = $7 * 65 = 455$

So, the total AQI of city D of remaining four days of the week = $455 - 201 = 254$

The average AQI of city D over the given three days = $201/3 = 67$

The average AQI of city D over the remaining four days = $254/4 = 63.5$

The required ratio = $67: 63.5 = 134: 127$

4. Questions

Let the AQI of city A on Monday be $100a$

On Monday,

The AQI's of city B = $100a * [1 - (20/100)] = 100a * (80/100) = 80a$

The AQI's of city C = $100a * [1 + (4/100)] = 100a * (104/100) = 104a$

The AQI's of city D = $100a * [1 - (8/100)] = 100a * (92/100) = 92a$

The AQI's of city E = $100a * [1 - (12/100)] = 100a * (88/100) = 88a$

The AQI's of city F = $100a * [1 + (12/100)] = 100a * (112/100) = 112a$

According to the question,

$$100a + 80a + 104a + 92a + 88a + 112a = 6 * 72$$

$$576a = 432$$

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Cities	AQI on Monday	AQI on Tuesday	AQI on Wednesday
A	$100 * 0.75 = 75$	—	—
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E	$88 * 0.75 = 66$	$66 + 6 = 72$	$90\% \text{ of } 72 = 64.8$
F	$112 * 0.75 = 84$	$84 + 4 = 88$	$105\% \text{ of } 88 = 92.4$

Answer: A

Since, the 2nd lowest AQI on Monday is 66, which is in city E.

And the highest AQI on Monday is 84, which is in city F.

So, M: N = 66: 84 = 11: 14

Statement I:

The value of $(M * N) = (11 * 14) = 154$

154 is not a perfect square.

So, Statement I is not true.

Statement II:

The value of $(15N - 5M) = (15 * 14) - (5 * 11) = 155$

155 is not a perfect cube.

So, statementII is not true.

Statement III:

The value of $(8M + 5N) = (8 * 11) + (5 * 14) = 158$

158 is neither a perfect square nor a perfect cube.

So, Statement III is not true.

Thus, none is true.

Hence, option A

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5. Questions

Let the AQI of city A on Monday be 100a

On Monday,

The AQI's of city B = $100a * [1 - (20/100)] = 100a * (80/100) = 80a$

The AQI's of city C = $100a * [1 + (4/100)] = 100a * (104/100) = 104a$

The AQI's of city D = $100a * [1 - (8/100)] = 100a * (92/100) = 92a$

The AQI's of city E = $100a * [1 - (12/100)] = 100a * (88/100) = 88a$

The AQI's of city F = $100a * [1 + (12/100)] = 100a * (112/100) = 112a$

According to the question,

$$100a + 80a + 104a + 92a + 88a + 112a = 6 * 72$$

$$576a = 432$$

$$a = 0.75$$

Cities	AQI on Monday	AQI on Tuesday	AQI on Wednesday
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E	$88 * 0.75 = 66$	$66 + 6 = 72$	90% of 72 = 64.8
F	$112 * 0.75 = 84$	$84 + 4 = 88$	105% of 88 = 92.4

Answer: E

Since, the original AQI of city A on Monday = 75

So, the new AQI of city A on Monday = $75 * [1 - (33(1/3)/100)] = 75 * (2/3) = 50$

The AQI of city D on Monday, now = $(92/100) * 50 = 46$

The AQI of city D on Tuesday, now = $46 - 9 = 37$

The AQI of city D on Wednesday, now = $(120/100) * 37 = 44.4$

The AQI of city E on Monday, now = $(88/100) * 50 = 44$

The AQI of city E on Tuesday, now = $44 + 6 = 50$

The AQI of city E on Wednesday, now = $(90/100) * 50 = 45$

The required percentage = $(44.4/45) * 100 = 98(2/3)\%$

6. Questions

Answer: D

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Quantity I:

The total AQI of all the given three days together in city C = $78 + 80 + 76 = 234$

The total AQI for the whole week = $75 * 7 = 525$

The total AQI over the remaining four days in city C = $525 - 234 = 291$

The average AQI over the remaining four days in city C = $291/4 = 72.75$

Quantity II:

The average AQI of cities B, C, D, E and F on Tuesday = $(50 + 80 + 60 + 72 + 88)/5 = 350/5 = 70$

So, Quantity I > Quantity II

Hence, option D

7. Questions

Let the initial investment of A be ₹x

So, the initial investment of B will be either ₹(x + 3000) or ₹(x - 3000).

Since, the ratio of the initial investment of B to that of C = 9: 5

So, the initial investment of C will be either ₹ $[5(x + 3000)/9]$ or ₹ $[5(x - 3000)/9]$.

Since, the initial investment of D is ₹4000 more than that of C.

So, the initial investment of D will be either ₹ $[5(x + 3000)/9 + 4000]$ or ₹ $[5(x - 3000)/9 + 4000]$.

The ratio of the investment of A to that of D = 15: 14

Case I:

$$x/[5(x + 3000)/9 + 4000] = 15/14$$

$$9x/(5x + 15000 + 36000) = 15/14$$

$$3x/(5x + 51000) = 5/14$$

$$42x = 25x + 255000$$

$$x = 255000/17$$

$$x = 15000$$

Case II:

$$x/[5(x - 3000)/9 + 4000] = 15/14$$

$$9x/(5x - 15000 + 36000) = 15/14$$

$$3x/(5x + 21000) = 5/14$$

$$42x = 25x + 105000$$

$$x = 105000/17$$

$$x = 6176.5 \text{ (approx.)}$$

Since, the initial investment of A is more than ₹7500.

So, case II can be neglected.

Now, the initial investment of A = ₹15000

The initial investment of B = $15000 + 3000 = ₹18000$

The initial investment of C = $18000 * (5/9) = ₹10000$

And the initial investment of D = $10000 + 4000 = ₹14000$

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Answer: E

Since, A's initial investment = ₹15000

And the B's initial investment = ₹18000

So, the ratio of A's profit share to B's profit share at the end of 8th month = $[(15000 * t) + (60\% \text{ of } 15000) * (8 - t)] : [18000 * (t + 2) + 66(2/3)\% \text{ of } 18000 * (8 - t - 2)]$

$$= [15000t + 72000 - 9000t] : [18000t + 36000 + 72000 - 12000t]$$

$$= [6000t + 72000] : [6000t + 108000]$$

$$= [6t + 72] : [6t + 108]$$

$$= [t + 12] : [t + 18]$$

Since, the ratio of A's profit share to B's profit share at the end of 8th month = 8: 11

So,

$$(t + 12)/(t + 18) = 8/11$$

$$11t + 132 = 8t + 144$$

$$3t = 12$$

$$t = 4$$

The time period for B withdrew $33(1/3)\%$ of his initial investment = $4 + 2 = 6$ months

8. Questions

Let the initial investment of A be ₹x

So, the initial investment of B will be either ₹(x + 3000) or ₹(x - 3000).

Since, the ratio of the initial investment of B to that of C = 9: 5

So, the initial investment of C will be either ₹[5(x + 3000)/9] or ₹[5(x - 3000)/9].

Since, the initial investment of D is ₹4000 more than that of C.

So, the initial investment of D will be either ₹[{5(x + 3000)/9} + 4000] or ₹[{5(x - 3000)/9} + 4000].

The ratio of the investment of A to that of D = 15: 14

Case I:

$$x/[{5(x + 3000)/9} + 4000] = 15/14$$

$$9x/(5x + 15000 + 36000) = 15/14$$

$$3x/(5x + 51000) = 5/14$$

$$42x = 25x + 255000$$

$$x = 255000/17$$

$$x = 15000$$

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Case II:

$$x/[\{5(x - 3000)/9\} + 4000] = 15/14$$

$$9x/(5x - 15000 + 36000) = 15/14$$

$$3x/(5x + 21000) = 5/14$$

$$42x = 25x + 105000$$

$$x = 105000/17$$

$$x = 6176.5 \text{ (approx.)}$$

Since, the initial investment of A is more than ₹7500.

So, case II can be neglected.

Now, the initial investment of A = ₹15000

The initial investment of B = 15000 + 3000 = ₹18000

The initial investment of C = 18000 * (5/9) = ₹10000

And the initial investment of D = 10000 + 4000 = ₹14000

Answer: C

Since, B's initial investment = ₹18000

And D's initial investment = ₹14000

So, M: N = 18000: 14000

M: N = 9: 7

Since, the total profit received by B and D together after 12 months = $[(M + N) * 5M * 100] = [(9 + 7) * 5 * 9 * 100] = ₹72000$

So, the difference between the profit shares of B and D = $72000 * (9 - 7)/(9 + 7) = 72000 * (2/16) = ₹9000$

9. Questions

Let the initial investment of A be ₹x

So, the initial investment of B will be either ₹(x + 3000) or ₹(x - 3000).

Since, the ratio of the initial investment of B to that of C = 9: 5 @QuizByMickey

So, the initial investment of C will be either ₹ $[5(x + 3000)/9]$ or ₹ $[5(x - 3000)/9]$.

Since, the initial investment of D is ₹4000 more than that of C.

So, the initial investment of D will be either ₹ $[\{5(x + 3000)/9\} + 4000]$ or ₹ $[\{5(x - 3000)/9\} + 4000]$.

The ratio of the investment of A to that of D = 15: 14

Case I:

$$x/[\{5(x + 3000)/9\} + 4000] = 15/14$$

$$9x/(5x + 15000 + 36000) = 15/14$$

$$3x/(5x + 51000) = 5/14$$

$$42x = 25x + 255000$$

$$x = 255000/17$$

$$x = 15000$$

Case II:

$$x/[\{5(x - 3000)/9\} + 4000] = 15/14$$

$$9x/(5x - 15000 + 36000) = 15/14$$

$$3x/(5x + 21000) = 5/14$$

$$42x = 25x + 105000$$

$$x = 105000/17$$

$$x = 6176.5 \text{ (approx.)}$$

Since, the initial investment of A is more than ₹7500.

So, case II can be neglected.

Now, the initial investment of A = ₹15000

The initial investment of B = 15000 + 3000 = ₹18000

The initial investment of C = 18000 * (5/9) = ₹10000

And the initial investment of D = 10000 + 4000 = ₹14000

Answer: B

The ratio of B's profit share after 4 months to C's profit share after 10 months = $(18000 * 4) : (10000 * 10)$
= 18: 25

So, the profit received by B after 4 months = $10500 * \{18/(25 - 18)\} = ₹27000$

And the profit received by C after 10 months = $10500 * 25/(25 - 18) = ₹37500$

The total SI received by B and C together after 2 years = $[(27000 + 37500) * 7.5 * 2]/100 = ₹9675$

10. Questions

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Let the initial investment of A be ₹x

So, the initial investment of B will be either ₹(x + 3000) or ₹(x - 3000).

Since, the ratio of the initial investment of B to that of C = 9: 5

So, the initial investment of C will be either ₹ $[5(x + 3000)/9]$ or ₹ $[5(x - 3000)/9]$.

Since, the initial investment of D is ₹4000 more than that of C.

So, the initial investment of D will be either ₹ $[\{5(x + 3000)/9\} + 4000]$ or ₹ $[\{5(x - 3000)/9\} + 4000]$.

The ratio of the investment of A to that of D = 15: 14

Case I:

$$x/[5(x + 3000)/9 + 4000] = 15/14$$

$$9x/(5x + 15000 + 36000) = 15/14$$

$$3x/(5x + 51000) = 5/14$$

$$42x = 25x + 255000$$

$$x = 255000/17$$

$$x = 15000$$

Case II:

$$x/[5(x - 3000)/9 + 4000] = 15/14$$

$$9x/(5x - 15000 + 36000) = 15/14$$

$$3x/(5x + 21000) = 5/14$$

$$42x = 25x + 105000$$

$$x = 105000/17$$

$$x = 6176.5 \text{ (approx.)}$$

Since, the initial investment of A is more than ₹7500.

So, case II can be neglected.

Now, the initial investment of A = ₹15000

The initial investment of B = 15000 + 3000 = ₹18000

The initial investment of C = 18000 * (5/9) = ₹10000

And the initial investment of D = 10000 + 4000 = ₹14000

Answer: B

Since, the time periods for the investments of A, B, C and D are 8 months, 10 months, 9 months and 15 months respectively.

So, the ratio between the profit shares of A, B, C and D = (15000 * 8): (18000 * 10): (10000 * 9): (14000 * 15)

$$= (3 * 8): (18 * 2): (2 * 9): (14 * 3)$$

$$= 4: 6: 3: 7$$

Since, the total profit received by A, B, C and D together = ₹35000

So, A's profit share = 35000 * (4/20) = ₹7000

B's profit share = 35000 * (6/20) = ₹10500

C's profit share = 35000 * (3/20) = ₹5250

And D's profit share = $35000 * (7/20) = ₹12250$

Statement I:

Since, the difference between the profit shares of B and C = $10500 - 5250 = ₹5250$

So, Statement I is not true.

Statement II:

The difference between the profit shares of B and D = $12250 - 10500 = ₹1750$

$$1750/140 = 12.5$$

Here, 1750 is not completely divisible by 140.

So, statement II is not true.

Statement III:

The total profit share of A and C together = $7000 + 5250 = ₹12250$

12250 is not a perfect square.

So, statement III is not true.

Thus, none is true.

Hence, option B

11. Questions

Trains	Speed (in m/s)	Length (in m)
A	X	$60(X - Y)$
B	$(X + 4)$	150
C	18	$10(X + 1)$
D	Y	$10(Y + 1)$
E	$(Y + 4)$	200

From Note I,

Formula: Distance = speed * Time

$$[60(X - Y) + 228] = X * 17$$

$$60X - 60Y + 228 = 17X$$

$$60Y - 43X = 228 \text{ -----}>(1)$$

From Note II,

$$10(Y + 1) = 10(X + 1) * (88/100)$$

$$Y + 1 = (X + 1) * (22/25)$$

$$25Y + 25 = 22X + 22$$

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$$22X - 25Y = 3$$

$$X = (3 + 25Y)/22 \text{ -----}>(2)$$

From equations (1) and (2):

$$60Y - 43 * [(3 + 25Y)/22] = 228$$

$$1320Y - 129 - 1075Y = 5016$$

$$245Y = 5145$$

$$Y = 21$$

From equation (2):

$$X = (3 + (25 * 21))/22$$

$$X = 528/22$$

$$X = 24$$

Trains	Speed (in m/s)	Length (in m)
A	24	$60 * (24 - 21) = 180$
B	$24 + 4 = 28$	150
C	18	$10 * (24 + 1) = 250$
D	21	$10 * (21 + 1) = 220$
E	$21 + 4 = 25$	200

Answer: C

We know that,

The speed of train A = 24 m/s

The speed of train B = 28 m/s

The length of train A = 180 m

The length of train B = 150 m

When the trains A and B are 450 m away from each other and coming towards each other, then the time taken, in which they will cross each other = $x = (180 + 150 + 450)/(24 + 28)$

$$x = 780/ 52$$

$$x = 15 \text{ seconds}$$

When the train B is 150 m behind the train A, then the time taken, in which the train B will cross the train A = $y = (180 + 150 + 150)/(28 - 24) = 480/4$

$$y = 120 \text{ seconds}$$

So, the required ratio = $x: y = 15: 120 = 1:8$

12. Questions

Trains	Speed (in m/s)	Length (in m)
A	X	$60(X - Y)$
B	$(X + 4)$	150
C	18	$10(X + 1)$
D	Y	$10(Y + 1)$
E	$(Y + 4)$	200

From Note I,

Formula: Distance = speed * Time

$$[60(X - Y) + 228] = X * 17$$

$$60X - 60Y + 228 = 17X$$

$$60Y - 43X = 228 \text{ -----}>(1)$$

From Note II,

$$10(Y + 1) = 10(X + 1) * (88/100)$$

$$Y + 1 = (X + 1) * (22/25)$$

$$25Y + 25 = 22X + 22$$

$$22X - 25Y = 3$$

$$X = (3 + 25Y)/22 \text{ -----}>(2)$$

From equations (1) and (2):

$$60Y - 43 * [(3 + 25Y)/22] = 228$$

$$1320Y - 129 - 1075Y = 5016$$

$$245Y = 5145$$

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From equation (2):

$$X = (3 + (25 * 21))/22$$

$$X = 528/22$$

$$X = 24$$

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Trains	Speed (in m/s)	Length (in m)
A	24	$60 * (24 - 21) = 180$
B	$24 + 4 = 28$	150
C	18	$10 * (24 + 1) = 250$
D	21	$10 * (21 + 1) = 220$
E	$21 + 4 = 25$	200

Answer: E

Since, the speed of train D = 21 m/s

And the length of train D = 220 m

So,

$$(220 + d) = 21 * 15$$

$$d = 315 - 220$$

$$d = 95 \text{ m}$$

Since, the speed of train E = 25 m/s

And the length of train E = 200 m

So,

$$(200 + 95 + 5) = 25 * t$$

$$300/25 = t$$

$$t = 12 \text{ seconds}$$

Since, the speed of train C = 18 m/s

And the length of train C = 250 m

So, the time taken by train C to cross a long tunnel of length $(2d - 8) \text{ m} = (250 + 2d - 8)/18$

$$= (250 + (2 * 95) - 8)/18$$

$$= 432/18 = 24 \text{ seconds}$$

Check option wise, here $t = 12$

a). $(t + 5) \text{ seconds} = 12 + 5 = 17$ is wrong

b). $'3t' \text{ seconds} = 3 * 12 = 36$ is wrong

c). $(t - 2) \text{ seconds} = 12 - 2 = 10$ is wrong

d). $(t + 10) \text{ seconds} = 12 + 10 = 22$ is wrong

e). $'2t' \text{ seconds} = 2 * 12 = 24$ is correct

Hence, option E

13. Questions

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Trains	Speed (in m/s)	Length (in m)
A	X	$60(X - Y)$
B	$(X + 4)$	150
C	18	$10(X + 1)$
D	Y	$10(Y + 1)$
E	$(Y + 4)$	200

From Note I,

Formula: Distance = speed * Time

$$[60(X - Y) + 228] = X * 17$$

$$60X - 60Y + 228 = 17X$$

$$60Y - 43X = 228 \text{ -----}>(1)$$

From Note II,

$$10(Y + 1) = 10(X + 1) * (88/100)$$

$$Y + 1 = (X + 1) * (22/25)$$

$$25Y + 25 = 22X + 22$$

$$22X - 25Y = 3$$

$$X = (3 + 25Y)/22 \text{ -----}>(2)$$

From equations (1) and (2):

$$60Y - 43 * [(3 + 25Y)/22] = 228$$

$$1320Y - 129 - 1075Y = 5016$$

$$245Y = 5145$$

$$Y = 21$$

From equation (2):

$$X = (3 + (25 * 21))/22$$

$$X = 528/22$$

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Trains	Speed (in m/s)	Length (in m)
A	24	$60 * (24 - 21) = 180$
B	$24 + 4 = 28$	150
C	18	$10 * (24 + 1) = 250$
D	21	$10 * (21 + 1) = 220$
E	$21 + 4 = 25$	200

Answer: E

From I:

The speed of train B = 28 m/s

The length of train B = 150 m

The required time taken = $(150 + 186)/28 = 336/28 = 12$ seconds

So, Statement I is not true.

From II:

The ratio of the length of train A to that of train E = $180:200 = 9:10$

So, statement II is true.

From III:

The average speed of trains A, C and D = $(24 + 18 + 21)/3 = 63/3 = 21$ m/s = $21 * (18/5) = 75.6$ km/h

So, statement III is true.

Therefore, only I

Hence, option E

14. Questions

Trains	Speed (in m/s)	Length (in m)
A	X	$60(X - Y)$
B	$(X + 4)$	150
C	18	$10(X + 1)$
D	Y	$10(Y + 1)$
E	$(Y + 4)$	200

From Note I,

Formula: Distance = speed * Time

$$[60(X - Y) + 228] = X * 17$$

$$60X - 60Y + 228 = 17X$$

$$60Y - 43X = 228 \text{ -----}>(1)$$

From Note II,

$$10(Y + 1) = 10(X + 1) * (88/100)$$

$$Y + 1 = (X + 1) * (22/25)$$

$$25Y + 25 = 22X + 22$$

$$22X - 25Y = 3$$

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$$X = (3 + 25Y)/22 \text{ -----}>(2)$$

From equations (1) and (2):

$$60Y - 43 * [(3 + 25Y)/22] = 228$$

$$1320Y - 129 - 1075Y = 5016$$

$$245Y = 5145$$

$$Y = 21$$

From equation (2):

$$X = (3 + (25 * 21))/22$$

$$X = 528/22$$

$$X = 24$$

Trains	Speed (in m/s)	Length (in m)
A	24	$60 * (24 - 21) = 180$
B	$24 + 4 = 28$	150
C	18	$10 * (24 + 1) = 250$
D	21	$10 * (21 + 1) = 220$
E	$21 + 4 = 25$	200

Answer: E

Since, the speed of train B = 28 m/s

The speed of train C = 18 m/s

The length of train B = 150 m

And the length of train C = 250 m

$$\text{So, } p = (150 + 250)/(28 - 18) = 400/10$$

$$p = 40 \text{ seconds}$$

Since, the speed of train D = 21 m/s

The speed of train E = 25 m/s

The length of train D = 220 m

And the length of train E = 200 m

$$\text{So, } q = (220 + 200)/(25 - 21) = 420/4$$

$$q = 105 \text{ seconds}$$

From option (a):

$$21p = 10q$$

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$$21 * 40 = 10 * 105$$

$$840 \neq 1050$$

So, option (a) is not correct.

From option (b):

$$2p + q = 185$$

$$(2 * 40) + 105 = 185$$

$$185 = 185$$

So, option (b) is correct.

From option (c):

$$p * q = 4200$$

$$40 * 105 = 4200$$

$$4200 = 4200$$

So, option (c) is correct.

Hence, both options (b) and (c) are correct.

15. Questions

Trains	Speed (in m/s)	Length (in m)
A	X	$60(X - Y)$
B	$(X + 4)$	150
C	18	$10(X + 1)$
D	Y	$10(Y + 1)$
E	$(Y + 4)$	200

From Note I,

Formula: Distance = speed * Time

$$[60(X - Y) + 228] = X * 17$$

$$60X - 60Y + 228 = 17X$$

$$60Y - 43X = 228 \text{ -----}>(1)$$

From Note II,

$$10(Y + 1) = 10(X + 1) * (88/100)$$

$$Y + 1 = (X + 1) * (22/25)$$

$$25Y + 25 = 22X + 22$$

$$22X - 25Y = 3$$

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$$X = (3 + 25Y)/22 \text{ -----}>(2)$$

From equations (1) and (2):

$$60Y - 43 * [(3 + 25Y)/22] = 228$$

$$1320Y - 129 - 1075Y = 5016$$

$$245Y = 5145$$

$$Y = 21$$

From equation (2):

$$X = (3 + (25 * 21))/22$$

$$X = 528/22$$

$$X = 24$$

Trains	Speed (in m/s)	Length (in m)
A	24	$60 * (24 - 21) = 180$
B	$24 + 4 = 28$	150
C	18	$10 * (24 + 1) = 250$
D	21	$10 * (21 + 1) = 220$
E	$21 + 4 = 25$	200

Answer: B

Since the average speed of trains C, D, E, and F = $72 \text{ km/h} = 72 * (5/18) = 20 \text{ m/s}$

So, the speed of train F = $(4 * 20) - (18 + 21 + 25) = 80 - 64 = 16 \text{ m/s}$

Since, the average length of trains A, B, C, and F = 175 m

So, the length of train F = $(4 * 175) - (180 + 150 + 250) = 700 - 580 = 120 \text{ m}$

And the time taken by train F to cross a long bridge whose length $232 \text{ m} = (120 + 232)/16 = 352/16 = 22 \text{ seconds}$

16. Questions

Answer: B

From equation P:

$$x^2 - 5x - 104 = 0$$

$$x^2 - 13x + 8x - 104 = 0$$

$$x(x - 13) + 8(x - 13) = 0$$

$$(x - 13)(x + 8) = 0$$

$$x = 13, -8$$

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From equation Q:

$$y^2 - 9y - 70 = 0$$

$$y^2 - 14y + 5y - 70 = 0$$

$$y(y - 14) + 5(y - 14) = 0$$

$$(y - 14)(y + 5) = 0$$

$$y = 14, -5$$

From equation R:

$$z^2 - 11z - 60 = 0$$

$$z^2 - 15z + 4z - 60 = 0$$

$$z(z - 15) + 4(z - 15) = 0$$

$$(z - 15)(z + 4) = 0$$

$$z = 15, -4$$

So, the correct combinations are P-II, Q-III, R-I

Hence, the correct answer is option B

17. Questions

Answer: E

From equation P:

$$4x^2 - 29x + 45 = 0$$

$$4x^2 - 20x - 9x + 45 = 0$$

$$4x(x - 5) - 9(x - 5) = 0$$

$$(x - 5)(4x - 9) = 0$$

$$x = 5, 9/4$$

From equation Q:

$$4y^2 + 7y - 36 = 0$$

$$4y^2 + 16y - 9y - 36 = 0$$

$$4y(y + 4) - 9(y + 4) = 0$$

$$(y + 4)(4y - 9) = 0$$

$$y = -4, 9/4$$

From equation R:

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$$z^2 - z - 20 = 0$$

$$z^2 - 5z + 4z - 20 = 0$$

$$z(z - 5) + 4(z - 5) = 0$$

$$(z - 5)(z + 4) = 0$$

$$z = 5, -4$$

So, the correct combinations are P-I, P-II, Q-I, Q-III, R-II, R-III

Hence, the correct answer is option E

18. Questions

Answer: C

From equation P:

$$3x^2 - 25x + 52 = 0$$

$$3x^2 - 12x - 13x + 52 = 0$$

$$3x(x - 4) - 13(x - 4) = 0$$

$$(x - 4)(3x - 13) = 0$$

$$x = 4, 13/3$$

From equation Q:

$$3y^2 - 8y - 16 = 0$$

$$3y^2 - 12y + 4y - 16 = 0$$

$$3y(y - 4) + 4(y - 4) = 0$$

$$(y - 4)(3y + 4) = 0$$

$$y = 4, -4/3$$

From equation R:

$$3z^2 + 8z - 16 = 0$$

$$3z^2 + 12z - 4z - 16 = 0$$

$$3z(z + 4) - 4(z + 4) = 0$$

$$(z + 4)(3z - 4) = 0$$

$$z = -4, 4/3$$

Thus, the correct combinations are P-I, R-III

So, the correct option is C.

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19. Questions

Answer: A

From equation P:

$$x^2 - x - 42 = 0$$

$$x^2 - 7x + 6x - 42 = 0$$

$$x(x - 7) + 6(x - 7) = 0$$

$$(x - 7)(x + 6) = 0$$

$$x = 7, -6$$

From equation Q:

$$y^2 - 25y + 136 = 0$$

$$y^2 - 17y - 8y + 136 = 0$$

$$y(y - 17) - 8(y - 17) = 0$$

$$(y - 17)(y - 8) = 0$$

$$y = 17, 8$$

From equation R:

$$z^2 - 11z - 102 = 0$$

$$z^2 - 17z + 6z - 102 = 0$$

$$z(z - 17) + 6(z - 17) = 0$$

$$(z - 17)(z + 6) = 0$$

$$z = 17, -6$$

Thus, the correct combinations are P-II, Q-III, R-I

So, the correct option is A.

20. Questions

Answer: C

From equation P:

$$2x^2 - 47x - 5A = 0$$

Since one root of 'x' is $(-3/2)$.

So,

$$2 * (-3/2)^2 - 47 * (-3/2) - 5A = 0$$

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$$(9/2) + (141/2) = 5A$$

$$5A = 75$$

$$A = 15$$

The value of A applies on equation P, we get

$$2x^2 - 47x - 5 * 15 = 0$$

$$2x^2 - 50x + 3x - 75 = 0$$

$$2x(x - 25) + 3(x - 25) = 0$$

$$(x - 25) (2x + 3) = 0$$

$$x = 25, -3/2$$

From equation Q:

$$2y^2 - 31y - 6A = 0$$

$$2y^2 - 31y - 6 * 15 = 0$$

$$2y^2 - 36y + 5y - 90 = 0$$

$$2y(y - 18) + 5(y - 18) = 0$$

$$(y - 18) (2y + 5) = 0$$

$$y = 18, -5/2$$

From equation R:

$$3z^2 - 67z - 8A = 0$$

$$3z^2 - 67z - 8 * 15 = 0$$

$$3z^2 - 72z + 5z - 120 = 0$$

$$3z(z - 24) + 5(z - 24) = 0$$

$$(z - 24) (3z + 5) = 0$$

$$z = 24, -5/3$$

So, the correct combinations are P-III, Q-I, R-II

Hence, the correct option is C.

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